Tail-record renewal at the deficit-four horizon

27 August 2026

Let

$$x_{t+1} = U_+(x_t) := \frac{3x_t + 1}{2^{a_t}}, \qquad a_t := v_2(3x_t + 1), \qquad A_t := \sum_{h=0}^{t-1} a_h, \tag{1}$$

be an infinite accelerated orbit of positive odd integers. Define its coefficient sequence and decoder depth by

$$Q_t := \frac{3^t}{2^{A_t}}, \qquad b_t := v_3(x_t + 1). \tag{2}$$

Assume throughout that $Q_t \to \infty$. A time j is a tail record if

$$Q_j < Q_t \qquad (t > j). \tag{3}$$

The tail-record times form an infinite sequence

$$j_0 < j_1 < j_2 < \cdots . \tag{4}$$

For a positive word $w = (c_0, \ldots, c_{r-1})$, put

$$\begin{aligned} A_0(w) &:= 0, \qquad A_h(w) := \sum_{t=0}^{h-1} c_t, \\ C_0(w) &:= 0, \quad C_{h+1}(w) := 3C_h(w) + 2^{A_h(w)}. \end{aligned} \tag{5}$$

and

$$D(w) := C_r(w) + 2^{A_r(w)}. \tag{6}$$

The affine identity is

$$2^{A_r(w)}(x_{j+r} + 1) = 3^r x_j + D(w) \tag{7}$$

whenever $w = (a_j, \ldots, a_{j+r-1})$.

Lemma 1 (geometry of consecutive tail records)**.** *Put* $r_i := j_{i+1} - j_i$ *and*

$$B_h := A_{j_i+h} - A_{j_i}, \qquad q_h := \frac{Q_{j_i+h}}{Q_{j_i}} = \frac{3^h}{2^{B_h}}. \tag{8}$$

Then

$$1 < q_{r_i} < q_h \qquad (1 \le h < r_i), \qquad a_{j_i} = a_{j_{i+1}} = 1. \tag{9}$$

If $r_i > 1$*, then* $a_{j_{i+1}-1} \ge 2$*.*

The consecutive-record gaps of length less than six are exactly

r_i	$(a_{j_i}, \dots, a_{j_{i+1}-1})$
1	1
2	12
4	1113 *or* 1122.

(10)

There are no consecutive-record gaps of length 3 *or* 5*.*

Proof. The values Q_t are pairwise distinct, since an equality $Q_u = Q_v$ with $u < v$ would give a positive power of 3 equal to a power of 2. Because $Q_t \to \infty$, every suffix has a unique minimum. The next tail record after j_i is the unique minimum of $\{Q_t : t > j_i\}$. This proves the first two inequalities in (9). At every tail record j, one has

$$\frac{Q_{j+1}}{Q_j} = \frac{3}{2^{a_j}} > 1,$$

so $a_j = 1$. If $r_i > 1$, then $j_{i+1} - 1$ is not a tail record, and (9) gives

$$\frac{Q_{j_{i+1}}}{Q_{j_{i+1}-1}} = \frac{3}{2^{a_{j_{i+1}-1}}} < 1.$$

Hence the last exponent is at least two.

It remains to prove (10). The endpoint condition $q_r < q_1 = 3/2$ and the strict barrier $q_r > 1$ give

$$1 < \frac{3^r}{2^{B_r}} < \frac{3}{2}. \tag{11}$$

For $r = 2$, this forces $B_2 = 3$, hence the word is 12. For $r = 3$, (11) forces $B_3 \ge 5$, while $q_3 > 1$ forces $B_3 \le 4$; hence no word exists. For $r = 4$, equation (11) forces $B_4 = 6$. The prefix barriers leave only

$$1113, \qquad 1122, \qquad 1212.$$

The last word is impossible because its second coefficient is $9/8$, less than its endpoint coefficient $81/64$. The first two words satisfy (9). For $r = 5$, equation (11) forces $B_5 \ge 8$, whereas $q_5 > 1$ forces $B_5 \le 7$. The case $r = 1$ was already proved. This establishes the table. □

Let

$$\mathcal{S}_4 = \{\varnothing, 12, 112, 1112, 1113, 1122, 1212, 11113\}. \tag{12}$$

A stem $s \in \mathcal{S}_4$ matches a positive odd integer x if it is the initial accelerated exponent word of x. Define

$$R_4(x) := \max_{\substack{s \in \mathcal{S}_4 \\ s \text{ matches } x}} \Big(|s| + v_2\big(U_+^{|s|}(x) + 1\big) - 1 \Big). \tag{13}$$

The empty stem makes the maximum well defined.

Lemma 2 (exact exponent-one tower)**.** *For every positive odd y, the exact initial run of exponent-one steps has length*

$$v_2(y+1) - 1. \tag{14}$$

Consequently, if a matching stem s contributes the value R to the maximum in (13), then the exponent prefix of length R is

$$s1^{R-|s|}. \tag{15}$$

Proof. Put $\nu = v_2(y+1)$. If $\nu \geq 2$, then $y \equiv -1 \pmod 4$, so $v_2(3y+1) = 1$, and

$$U_+(y) + 1 = \frac{3(y+1)}{2}. \tag{16}$$

Thus one exponent-one step lowers ν by exactly one. When $\nu = 1$, one has $y \equiv 1 \pmod 4$, so the next exponent is at least two. This proves both assertions. □

Theorem 3 (deficit-four record-renewal dichotomy)**.** *For each i, put*

$$r_i := j_{i+1} - j_i, \qquad R_i := R_4(x_{j_i}). \tag{17}$$

Exactly one of the following alternatives holds.

1. *If $r_i > R_i$, then*
$$r_i \geq 6, \qquad b_{j_{i+1}} \leq r_i - 5. \tag{18}$$
2. *If $r_i \leq R_i$, then the record-gap word p_i belongs to*
$$\mathcal{A}_4 := \{1, 12, 1113, 1122\}, \tag{19}$$
and
$$R_{i+1} \geq R_i - r_i. \tag{20}$$

Its decoder transport is exact:

p_i	r_i	$A_{r_i}(p_i)$	$b_{j_{i+1}}$
1	1	1	$b_{j_i}+1$
12	2	3	0
1113	4	6	1
1122	4	6	0.

(21)

Proof. Fix i and suppose first that $r_i \le R_i$. Choose a stem s that attains the maximum R_i in (13). By Lemma 2, the prefix through time R_i is s followed by exponent-one steps. Since j_{i+1} is the unique minimum of the coefficient sequence after j_i, its relative coefficient is smaller than that at every other positive time through R_i.

The required finite comparison is

s	p_i	stem remaining after p_i
$\varnothing$	1	$\varnothing$
12	12	$\varnothing$
112	1	12
1112	1	112
1113	1113	$\varnothing$
1122	1122	$\varnothing$
1212	12	12
11113	1	1113.

(22)

Indeed, the relative coefficient vectors through the nonempty stems are

$$\begin{aligned}
12: &\quad (3/2, 9/8),\\
112: &\quad (3/2, 9/4, 27/16),\\
1112: &\quad (3/2, 9/4, 27/8, 81/32),\\
1113: &\quad (3/2, 9/4, 27/8, 81/64),\\
1122: &\quad (3/2, 9/4, 27/16, 81/64),\\
1212: &\quad (3/2, 9/8, 27/16, 81/64),\\
11113: &\quad (3/2, 9/4, 27/8, 81/16, 243/128).
\end{aligned} \tag{23}$$

The unique minimum in each row occurs at the boundary displayed in (22). Every appended exponent-one step multiplies the coefficient by $3/2$, so it creates no new minimum. This proves (19). The remaining word through time R_i is again a stem from $\mathcal{S}_4$ followed by its exact exponent-one tower. Applied at $x_{j_{i+1}}$, definition (13) therefore gives (20).

Direct use of (5)–(6) gives

p	1	12	1113	1122
$D(p)$	3	13	129	137
$v_3(D(p))$	1	0	1	0.

(24)

For $p = 1$, equation (7) reduces to

$$2(x_{j_i+1} + 1) = 3(x_{j_i} + 1),$$

which gives the first decoder rule in (21). For the other three words, $v_3(D(p)) < |p|$, while the first term on the right of (7) has valuation at least $|p|$. Hence the valuation of the sum is $v_3(D(p))$, proving the remaining rules.

Now suppose that $r_i > R_i$. Lemma 1 and the definition (13) show that none of the gaps in (10) can occur: each of their words is a matching stem and therefore gives $R_i \geq r_i$. Thus $r_i \geq 6$. The block from x_{j_i} is a strict barrier because j_i is a tail record. The proved fixed-start deficit-four horizon theorem, applied with $r_i \geq 4$ and $r_i > R_i$, gives the second assertion in (18). □

Corollary 4 (record-renewal trichotomy)**.** *Every orbit satisfying the hypotheses has at least one of the following three properties.*

1. *Infinitely many consecutive tail-record transitions satisfy $r_i > R_i$.*
2. *From some tail record onward every transition lies inside its current horizon, and the resulting word over $\mathcal{A}_4$ contains arbitrarily long consecutive runs of the one-letter atom* 1*.*
3. *From some tail record onward every transition lies inside its current horizon, and the decoder depth b_t is bounded on that tail.*

In the third alternative, if every run of the atom 1 *has length at most M, then one may take*

$$b_t \leq M + 4 \tag{25}$$

throughout a sufficiently late tail.

Proof. If the first alternative fails, Theorem 3 shows that all sufficiently late record gaps belong to $\mathcal{A}_4$. The three nontrivial atoms occur infinitely often. Otherwise the accelerated exponent word would eventually be all ones, contradicting the finite exact run length in Lemma 2.

If the intervening runs of the atom 1 are unbounded, the second alternative holds. Otherwise let M bound their lengths. Equation (21) shows that each nontrivial atom resets the decoder depth to zero or one, whereas each atom 1 raises it by one. Inside a nontrivial atom there are at most three initial exponent-one steps before the displayed reset. Consequently $b_t \le M + 4$ on the resulting tail, proving the third alternative and (25). □

Theorem 5 (record carry and explicit coalescing ancestor)**.** *Fix consecutive tail records $j_i < j_{i+1}$, put*

$$r := j_{i+1} - j_i > 1, \qquad x := x_{j_i}, \qquad y := x_{j_{i+1}}, \qquad B := A_{j_{i+1}} - A_{j_i}, \tag{26}$$

and let $w = (a_{j_i}, \dots, a_{j_{i+1}-1})$. With $\alpha := \log_2 3$ and $\{u\} := u - \lfloor u \rfloor$, one has

$$B = \lfloor r\alpha \rfloor, \qquad 0 < \{r\alpha\} < \alpha - 1, \qquad \frac{3^r}{2^B} = 2^{\{r\alpha\}} \in (1, 3/2). \tag{27}$$

Moreover,

$$\frac{C(w)}{2^B} = \frac{1}{3} \sum_{h=0}^{r-1} \frac{q_r}{q_h} < \frac{r + 1/2}{3}, \tag{28}$$

where the q_h are defined in (8), and consequently

$$\frac{3^r}{2^B} x < y < \frac{3^r}{2^B} x + \frac{r + 1/2}{3} < \frac{3}{2} x + \frac{r + 1/2}{3}. \tag{29}$$

Put $b := v_3(y + 1)$. Then

$$b \ge 1 \quad \Longleftrightarrow \quad a_{j_{i+1}-1} \text{ is odd.} \tag{30}$$

If these equivalent conditions hold, the positive odd integer

$$z := 2^b \frac{y + 1}{3^b} - 1 \tag{31}$$

satisfies

$$U_+^b(z) = y, \tag{32}$$

and all b exponents on this coalescing path are one. It is smaller than the record start x exactly when

$$\left(\frac{2}{3}\right)^b (y + 1) < x + 1. \tag{33}$$

In particular, either of the following explicit inequalities is sufficient:

$$\left(\frac{2}{3}\right)^b \left(\frac{3}{2}x + \frac{r+1/2}{3} + 1\right) < x + 1, \tag{34}$$

or

$$b \geq 2 \quad \textit{and} \quad 4r < 9x + 13. \tag{35}$$

Proof. Lemma 1 gives

$$1 < \frac{3^r}{2^B} < \frac{3}{2}.$$

Taking base-two logarithms gives

$$0 < r\alpha - B < \alpha - 1 < 1.$$

Since B is an integer, this proves (27).

Expansion of the carry recurrence in (5) gives

$$C(w) = \sum_{h=0}^{r-1} 3^{r-1-h} 2^{B_h}.$$

After division by 2^B, each summand becomes

$$\frac{1}{3}\frac{3^{r-h}}{2^{B-B_h}} = \frac{1}{3}\frac{q_r}{q_h}.$$

For $h = 0$, Lemma 1 gives $q_r/q_0 = q_r < 3/2$. For $1 \leq h < r$, it gives $q_r/q_h < 1$. Summation proves (28). The affine orbit identity

$$y = \frac{3^r}{2^B}x + \frac{C(w)}{2^B}$$

then proves (29).

Write $a := a_{j_{i+1}-1}$. The final orbit step gives

$$2^a y \equiv 1 \pmod 3.$$

Thus $y \equiv (-1)^a \pmod 3$. Hence $3 \mid y + 1$ exactly when a is odd, which proves (30).

Suppose $b \geq 1$. Formula (31) defines a positive odd integer. For $0 \leq h \leq b$, put

$$z_h + 1 := 2^{b-h}\frac{y+1}{3^{b-h}}. \tag{36}$$

Every z_h is a positive odd integer, $z_0 = z$, and $z_b = y$. Moreover,

$$z_{h+1} + 1 = \frac{3(z_h + 1)}{2}. \tag{37}$$

Since j_{i+1} is a tail record, Lemma 1 gives $a_{j_{i+1}} = 1$, and therefore $v_2(y+1) \geq 2$. It follows from (36) that $v_2(z_h+1) \geq 2$ for $h < b$. Lemma 2 and (37) now show that each step from z_h to z_{h+1} has exact exponent one. This proves (32).

Equation (31) gives

$$z + 1 = \left(\frac{2}{3}\right)^b (y + 1),$$

which proves (33). Combining this identity with (29) proves (34). If $b \geq 2$, the left side of (34) is at most

$$\frac{4}{9}\left(\frac{3}{2}x + \frac{r + 1/2}{3} + 1\right) = \frac{2}{3}x + \frac{4r + 14}{27}.$$

This is less than $x + 1$ whenever $4r < 9x + 13$, proving (35). □

Theorem 6 (weighted monotonicity of decoder ancestors)**.** *Suppose that the transition from j_i to j_{i+1} lies inside the current deficit-four horizon. Put*

$$\mathcal{W}_i := 2^{b_{j_i}} \frac{x_{j_i} + 1}{3^{b_{j_i}}}, \qquad \kappa(b) := \begin{cases} 1, & b = 0, \\ 32/27, & b \geq 1. \end{cases} \tag{38}$$

Then $\mathcal{W}_i - 1$ is a positive all-one coalescing ancestor of x_{j_i}, and the four interior record atoms act by

p_i	$b_{j_{i+1}}$	$\mathcal{W}_{i+1}$
1	$b_{j_i} + 1$	$\mathcal{W}_i$
12	0	$\frac{9}{8}\left(\frac{3}{2}\right)^{b_{j_i}} \mathcal{W}_i + \frac{1}{2}$
1113	1	$\frac{27}{32}\left(\frac{3}{2}\right)^{b_{j_i}} \mathcal{W}_i + \frac{1}{2}$
1122	0	$\frac{81}{64}\left(\frac{3}{2}\right)^{b_{j_i}} \mathcal{W}_i + \frac{7}{8}.$

(39)

In every case,

$$\boxed{\kappa(b_{j_{i+1}})\mathcal{W}_{i+1} \geq \kappa(b_{j_i})\mathcal{W}_i.} \tag{40}$$

Consequently, along any finite or infinite chain of interior record renewals,

$$\mathcal{W}_k \geq \frac{27}{32}\mathcal{W}_i \qquad (k \geq i). \tag{41}$$

In particular, every run of the one-letter atom 1 *leaves the canonical coalescing ancestor exactly unchanged; unbounded exponent-one towers do not produce cumulative ancestor descent.*

Proof. For any positive odd state v, let

$$b := v_3(v+1), \qquad \mathcal{W}(v) := 2^b \frac{v+1}{3^b}.$$

The construction in the proof of Theorem 5 shows that $\mathcal{W}(v) - 1$ reaches v through exactly b exponent-one steps. This proves the first assertion.

Write $Y := x_{j_i} + 1$, $Y' := x_{j_{i+1}} + 1$, and $b := b_{j_i}$. Equations (7) and (24) give, for the four atoms,

p_i	block equation
1	$2Y' = 3Y$
12	$8Y' = 9Y + 4$
1113	$64Y' = 81Y + 48$
1122	$64Y' = 81Y + 56.$

(42)

Since

$$Y = \left(\frac{3}{2}\right)^b \mathcal{W}_i, \tag{43}$$

substitution into (42), together with the decoder endpoints in (21), proves every formula in (39). In particular, the factor $3/2$ in the one-letter block raises the decoder depth at the same time as it changes Y, so $\mathcal{W}_{i+1} = \mathcal{W}_i$.

It remains to verify (40). For the atom 1, the weight κ increases from 1 to $32/27$ when $b = 0$ and is otherwise unchanged. For the atom 12, the leading multiplier in (39) is at least $9/8$ when $b = 0$ and at least $27/16 > 32/27$ when $b \geq 1$. For the atom 1113, multiplication by the terminal weight gives the leading factor

$$\frac{32}{27}\frac{27}{32}\left(\frac{3}{2}\right)^b = \left(\frac{3}{2}\right)^b.$$

This equals the initial weight when $b = 0$ and exceeds $32/27$ when $b \geq 1$. Finally, the leading multiplier for 1122 is at least $81/64 > 1$ when $b = 0$ and at least $243/128 > 32/27$ when $b \geq 1$. All additive terms in (39) are positive, so (40) follows. Iteration and $1 \leq \kappa(b) \leq 32/27$ give (41). □

Theorem 7 (escape of canonical decoder ancestors)**.** *Let $(x_t)_{t\geq 0}$ be an injective infinite accelerated orbit. For every t, put*

$$b_t := v_3(x_t + 1), \qquad \mathcal{W}_t := 2^{b_t}\frac{x_t + 1}{3^{b_t}}. \tag{44}$$

Then

$$\boxed{\mathcal{W}_t \longrightarrow \infty.} \tag{45}$$

More precisely, for every positive integer M,

$$\#\{t \geq 0 : \mathcal{W}_t = M\} \leq v_2(M). \tag{46}$$

The left side is zero when M is odd.

Suppose additionally that $x_0 = n$ is the least positive odd seed with an unbounded accelerated orbit. Then

$$\mathcal{W}_t \geq n + 1 \qquad (t \geq 0), \tag{47}$$

with equality only during the exact initial exponent-one tower of n. In particular, after that finite tower every canonical decoder ancestor is strictly larger than n.

Proof. Every $\mathcal{W}_t$ is even. Indeed, if $b_t = 0$, then $\mathcal{W}_t = x_t + 1$ is even, while the assertion is immediate when $b_t \geq 1$.

Fix an even integer M and put $z := M - 1$. If $\mathcal{W}_t = M$, the construction in Theorem 5 gives

$$U_+^{b_t}(z) = x_t, \tag{48}$$

and all b_t exponents on this path are one. Lemma 2 therefore gives

$$0 \leq b_t \leq v_2(M) - 1. \tag{49}$$

For a fixed value of b_t, equation (48) determines x_t. Injectivity of the orbit makes the values x_t pairwise distinct, so each integer in the range (49) occurs for at most one t. This proves (46).

For any fixed H, only finitely many even integers $M \leq H$ exist, and each occurs only finitely often by (46). Hence $\mathcal{W}_t > H$ for all sufficiently large t. Since H is arbitrary, (45) follows.

Finally, the orbit of $\mathcal{W}_t - 1$ reaches the unbounded state x_t. It is therefore unbounded, so global leastness of n gives (47). If we take $t = 0$ in (47), the defining formula (44) first shows that $b_0 = 0$ and $\mathcal{W}_0 = n + 1$. If equality holds at any t, then (48) becomes

$$U_+^{b_t}(n) = x_t = U_+^t(n).$$

Injectivity gives $b_t = t$. The coalescing path in (48) then shows that the first t exponents of n are all one. Conversely, every state in the exact initial exponent-one tower has canonical ancestor n. Lemma 2 shows that this tower is finite, completing the proof. □

Theorem 8 (two-prime stationary budget at coefficient records)**.** *Assume in addition that the normalized height has a finite limit*

$$\mathcal{K} := \lim_{t\to\infty} \frac{x_t}{Q_t} < \infty. \tag{50}$$

Put

$$\nu_t := v_2(x_t + 1), \qquad G_t := \frac{x_t + 1}{Q_t}, \qquad \gamma := \log_3 \frac{3}{2}. \tag{51}$$

Then, for every t,

$$\boxed{2^{\nu_t} 3^{b_t} \le x_t + 1 = Q_t G_t \le \mathcal{K} Q_t.} \tag{52}$$

Consequently,

$$\boxed{\mathcal{W}_t \ge G_0 \left(\frac{2}{\mathcal{K}}\right)^{\gamma} Q_t^{\log_3 2}.} \tag{53}$$

In particular, $Q_t \to \infty$ quantitatively forces $\mathcal{W}_t \to \infty$ and

$$\frac{Q_t}{(3/2)^{b_t}} \longrightarrow \infty. \tag{54}$$

Let

$$\alpha := \log_2 3, \qquad \ell_i := \nu_{j_i} - 1. \tag{55}$$

Thus ℓ_i is the exact initial exponent-one run at the ith tail record. Then

$$\boxed{\ell_i + 1 + \alpha b_{j_i} \le (\alpha - 1) i + \log_2(\mathcal{K} Q_{j_0}).} \tag{56}$$

Since $\ell_i \ge 1$, this gives the explicit record-index ceiling

$$\boxed{b_{j_i} \le \frac{\alpha - 1}{\alpha}, i + \frac{\log_2(\mathcal{K} Q_{j_0}) - 2}{\alpha}.} \tag{57}$$

Hence $i - b_{j_i} \to \infty$.

Finally, let L_i be the number of consecutive one-letter record atoms beginning at j_i. This number is finite and

$$\boxed{L_i + 2 + \alpha b_{j_i} \le (\alpha - 1) i + \log_2(\mathcal{K} Q_{j_0}).} \tag{58}$$

Thus the two interior mechanisms consume the same stationary budget: decoder depth has asymptotic record-index slope at most $(\alpha - 1)/\alpha$, while an exponent-one record tower has slope at most $\alpha - 1$.

Proof. The coprime powers 2^{ν_t} and 3^{b_t} both divide $x_t + 1$, so

$$2^{\nu_t} 3^{b_t} \mid x_t + 1. \tag{59}$$

To prove the upper bound in (52), first note that

$$G_{t+1} - G_t = \frac{2^{a_t} - 2}{3Q_t} \geq 0. \tag{60}$$

Moreover, $G_t = x_t/Q_t + 1/Q_t \to \mathcal{K}$. Hence $G_t \leq \mathcal{K}$, and (52) follows from (59).

Since $\nu_t \geq 1$, equation (52) gives

$$3^{b_t} \leq \frac{\mathcal{K} Q_t}{2}. \tag{61}$$

Raising (61) to the power $\gamma = \log_3(3/2)$ yields

$$\left(\frac{3}{2}\right)^{b_t} \leq \left(\frac{\mathcal{K} Q_t}{2}\right)^{\gamma}. \tag{62}$$

On the other hand, (44) and (51) give

$$\mathcal{W}_t = \frac{Q_t G_t}{(3/2)^{b_t}}. \tag{63}$$

The monotonicity in (60) gives $G_t \geq G_0$. Substitution of (62) into (63), together with $1 - \gamma = \log_3 2$, proves (53). Equations (53) and (63) prove (54).

By Lemma 1,

$$\frac{Q_{j_{i+1}}}{Q_{j_i}} \leq \frac{3}{2},$$

and therefore

$$Q_{j_i} \leq Q_{j_0} \left(\frac{3}{2}\right)^{i}. \tag{64}$$

Apply (52) at j_i and take binary logarithms. Since $\nu_{j_i} = \ell_i + 1$, equation (64) gives (56). Every tail record begins with an exponent-one step, so $\ell_i \geq 1$; this proves (57), and its linear coefficient is strictly less than one.

It remains to prove (58). An infinite sequence of consecutive one-letter atoms would give an infinite exponent-one run, contrary to Lemma 2. Thus L_i is finite. The L_i one-letter gaps are followed by another record gap, whose first exponent is again one by Lemma 1. Hence $\ell_i \geq L_i + 1$. Substitution in (56) proves (58). □